Understanding Bag-of-Words and corpora.Dictionary in Python

Discover how the Bag-of-Words model can be applied and visualised in a Python project.

The Bag-of-Words (BoW) model is a fundamental technique for text processing and natural language processing (NLP). It considers each word as an integer number and assigns the frequency of its occurrences within a document constituting the corpus. In Python, the Gensim library provides powerful tools to work with BoW models, including the 'corpora.Dictionary' class. In rule-based NLP, BoW is an intermediate stage where it receives data from the preliminary text processing steps like cleaning, removal of stopwords and punctuations, and so on, and then provides this data for high-level analysis to understand the corpus. Let's explore how the BoW approach is applied and visualised in a Python project.

The corpora.Dictionary class in Gensim is a mapping between words and their integer IDs. It helps to create the BoW representation of text documents, which is essential for many NLP tasks, including topic modelling.

You can create a dictionary as follows:

```
import numpy as np
from gensim import corpora
#Sample preprocessed texts
texts = [["PHI", "EWP","PHI"],["EWP","Elsevier"],["PHI","Else
vier","EWP"]]
dictionary = corpora.Dictionary(texts)
#Display the dictionary **
print(dictionary.token2id)
## {'EWP': 0, 'PHI': 1, 'Elsevier': 2}
```

Creating the Bag-of-Words (BoW) model

The BoW model represents each document as a list of word IDs and their corresponding frequencies. This can be achieved using the *doc2bow()* method. Here's an example:

```
bow_corpus = [dictionary.doc2bow(text) for text in texts]
print(bow_corpus)
## [[(0, 1), (1, 2)], [(0, 1), (2, 1)], [(0, 1), (1, 1), (2,
1)]]
```

This bag of words can be easily understood by using a for-next-loop to see each word's document-wise frequency:

```
# Print corpus
for i, doc in enumerate(bow_corpus):
    print(f"Document {i+1}: {doc}")
## Document 1: [(0, 1), (1, 2)]
## Document 2: [(0, 1), (2, 1)]
## Document 3: [(0, 1), (1, 1), (2, 1)]
```

To process a corpus, the most important step is to have a list of words of the corpus vocabulary, as corpora. Dictionary() creates a dictionary, i.e., unique (key, value) pairs from the corpus.

```
#Corpus Dictionary
vocabulary = [dictionary[id] for id in dictionary.keys()]
print(vocabulary)
## ['EWP', 'PHI', 'Elsevier']
```

This vocabulary-frequency extraction from a corpus is